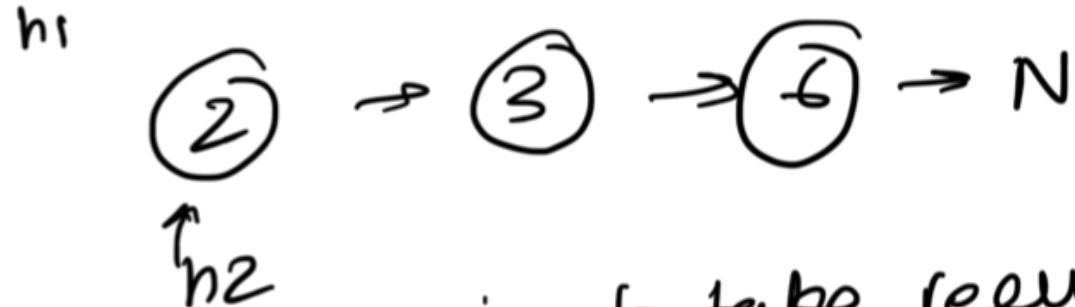
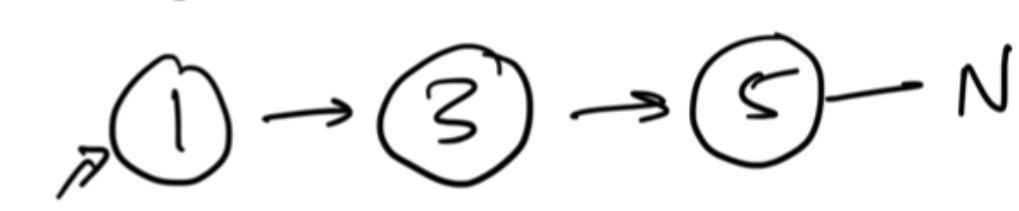


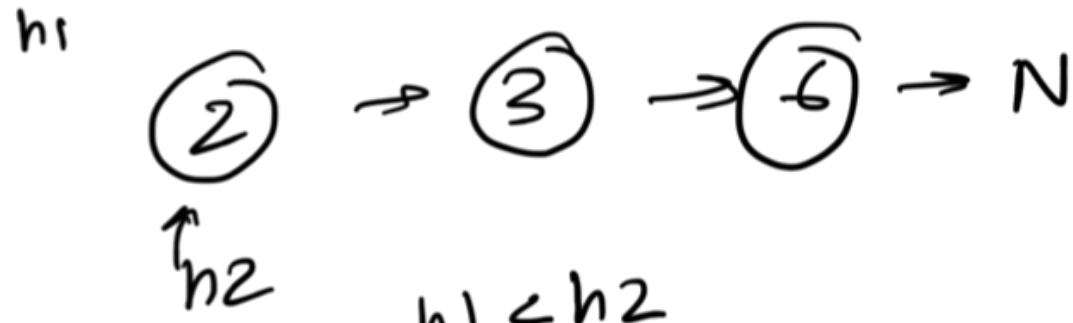
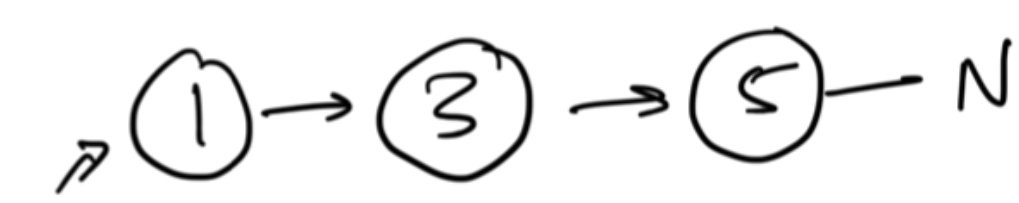
Merge 2 sorted lists



going to take recursive approach

```
ListNode * merge(h1, h2) {
    if (h1 == null || h2 == null) {
        return h2; // if h1 is null, return h2
        return h1; // if h2 is null, return h1
    }
    if (h1->val <= h2->val) {
        h1->next = merge(h1->next, h2);
        return h1;
    }
    else {
        h2->next = merge(h1, h2->next);
        return h2;
    }
}
```

TC: $O(n+m)$



$h_1 < h_2$

